Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/05/2022
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Does Intermittent or Continuous Artificial Gravity Counteract Long Duration Bed Rest Induced Neurocognitive Declines?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rachaelseidler@ufl.edu	**Fax:**	FY
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City:	Gainesville	**State:**	FL
Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-AGBR. Appendix G: Physiological & Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest
Start Date:	07/01/2018	**End Date:**	12/31/2022
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2022 per L. Juliette/JSC (Ed., 6/2/22). NOTE: End date changed to 6/30/2022 per NSSC information (Ed., 6/18/21) NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 8/28/20)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Mulavara, Ajitkumar Ph.D. (KBR/NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC18K0783		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Research Announcement NNJ15ZSA001N-AGBR: Appendix G: Physiological and Behavioral Responses in Humans to Intermittent Artificial Gravity during Bed Rest invites proposals to investigate whether 30 minutes of artificial gravity (AG) per day, delivered either continuously or in six, five minute intermittent blocks, mitigates changes that occur in a spaceflight analog environment. We have recently completed a 70 day 6° head down tilt bed rest study in which we identified specific bed rest induced changes in brain structure, brain function, sensorimotor, and cognitive performance. Here, we select those that are sensitive to this spaceflight analog environment as outcome metrics for the following aims: Aim 1) Determine whether 30 minutes per day of supine centrifugation +1Gz at the heart is effective at mitigating neurocognitive declines induced by 60 days of 6° head down tilt bed rest in comparison to a control group that is not exposed to AG. Aim 2) Determine whether centrifugation is more effective and better tolerated as a countermeasure when it is delivered continuously for 30 minutes or in six, five minute sessions separated by breaks of five minutes each. Our approach utilizes cutting edge neuroimaging techniques and a broad ranging battery of sensory, motor, and cognitive assessments that we have found to be sensitive indices of bed rest—and in several cases—spaceflight induced performance decrements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research on adaptive plasticity associated with behavioral training has greatly expanded over the past two decades, but research on maladaptive plasticity occurring with immobilization is scant. A greater understanding of brain structural and functional changes, and the concomitant behavioral effects resulting from limb disuse and unloading has implications for rehabilitation of those immobilized by injury, disease, or even simple inactivity.
Task Progress:	This project was conducted as a part of the Artificial Gravity Bed Rest – European Space Agency ("AGBRESA") campaign at :envihab in Cologne, Germany. This was a 60 day head-down tilt bed rest (HDBR) study with daily centrifugation as a potential countermeasure. One group did not participate in centrifugation, one received 30 minutes per day of continuous centrifugation, and one received 30 minutes per day of intermittent centrifugation in 5 minute sessions. The data were collected at :envihab, with data collection complete in December, 2019. Data transfer from Germany to my laboratory was complete in March, 2020. We had an initial slowing of data analyses due to the COVID pandemic. Therefore, we are still in the midst of manuscript and report preparation. Our investigation specifically focused on whether 30 minutes of daily artificial gravity (AG) is sufficient for mitigating bed rest induced changes in brain structure, function, and behavior. One group received 30 minutes of continuous AG (cAG) whereas another received 30 minutes per day in six intermittent bouts (iAG). We have analyzed the data and published the results for the behavioral assessments. We observed no significant differences between the three groups in terms of HDBR-associated changes in cognition, balance, and functional mobility. Although the effect was not statistically significant (perhaps due to the small group sample sizes), those that underwent artificial gravity had a 10 point advantage in balance equilibrium scores following bed rest than those that did not. Interestingly, the intermittent AG group reported less severe motion sickness symptoms than the continuous group during centrifugation; iAG motion sickness levels were not elevated above those of controls who did not undergo AG. They also had a shorter duration of post-AG illusory motion than cAG. Moreover, the two AG groups performed the paced auditory serial addition test weekly while undergoing AG; their performance was more accurate than that of controls, who performed the test while in HDBR. Although AG did not counteract HDBR-induced gait and balance declines, iAG did not cause motion sickness and was associated with better self-motion perception during AG ramp-down. Additionally, both AG groups had superior cognitive performance while undergoing AG relative to controls; this may reflect attention or motivation differences between the groups. [Ed. Note: See Cumulative Bibliography.] These behavioral findings have been published in: Tays GD, McGregor HR, Lee JK, Beltran N, Kofman IS, De Dios YE, Mulder E, Bloomberg JJ, Mulavara AP, Wood SJ, Seidler RD. "The effects of 30 minutes of artificial gravity on cognitive and sensorimotor performance in a spaceflight analog environment. " Front. Neural Circuits. 2022 March 2;16:784280. https://
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Tays GD, McGregor H R, Lee JK, Beltran N, Kofman IS, De Dios YE, Mulder E, Bloomberg JJ, Mulavara AP, Wood SJ, Seidler RD. "The effects of 30 minutes of artificial gravity on cognitive and sensorimotor performance in a spaceflight analog environment. " Front. Neural Circuits. 2022 March 2;16:784280. https://doi.org/10.3389/fncir.2022.784280 PMID: 35310547; PMCID: PMC892404 , Mar-2022